

Data Cleaning: From Theory to Practice

A comprehensive guide to discrepancy detection and metadata analysis in the data cleaning process

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The Data Cleaning Challenge

Missing values, noise, and inconsistencies contribute to inaccurate data that can derail analysis and lead to faulty conclusions. So far, we've explored techniques for handling missing data and for smoothing data. But data cleaning is a big job requiring systematic approaches.

The question remains: How exactly does one proceed in tackling this complex task? Are there any tools and methodologies to help streamline the process?

The Process Begins

Discrepancy detection is the first step in data cleaning as a process. Understanding the sources and patterns of data quality issues is essential before applying corrective measures.

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What Causes Data Discrepancies?

Discrepancies in datasets arise from multiple sources, each requiring different detection and correction strategies. Understanding these root causes helps analysts anticipate common problems and design appropriate validation rules.



Poorly Designed Forms

Data entry forms with many optional fields create gaps and inconsistencies in collected data



Human Error

Mistakes during manual data entry introduce typos, incorrect values, and formatting issues



Deliberate Errors

Respondents may withhold information or provide inaccurate data to protect privacy



Data Decay

Outdated information such as old addresses and expired contact details lose relevance over time

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Technical Sources of Data Errors

Inconsistent Representations

Different formats and coding schemes used across systems create integration challenges

- Varying date formats (MM/DD/YYYY vs DD/MM/YYYY)
- Inconsistent use of abbreviations and codes
- Different naming conventions for attributes

Instrumentation Errors

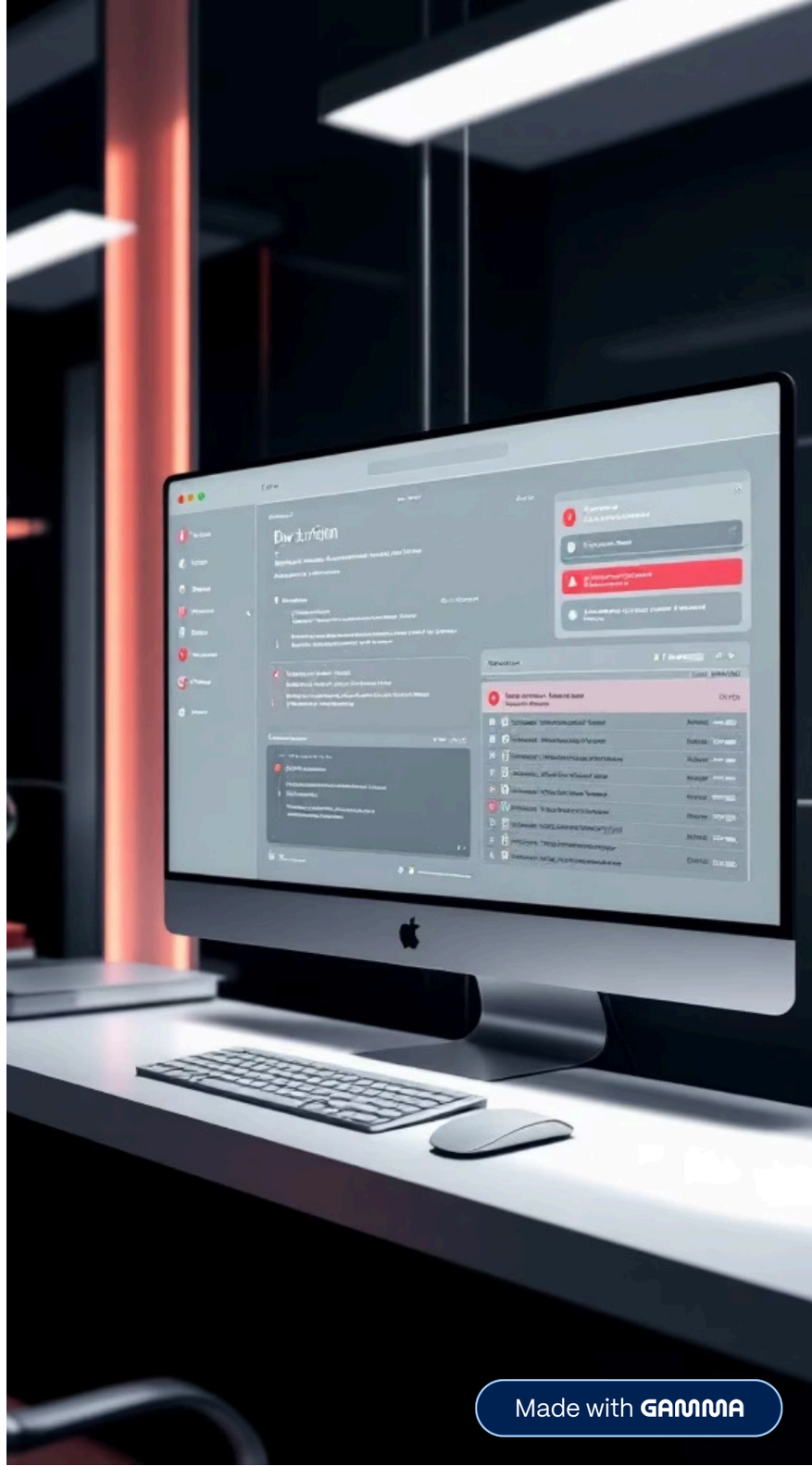
Device malfunctions and system errors during automated data collection introduce noise

- Sensor calibration issues
- System crashes during recording
- Transmission errors

Data Integration Issues

Merging datasets from different sources reveals mismatches in structure and content

- Same attribute with different names
- Conflicting value domains
- Duplicate records with variations



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Discrepancy Detection: The Starting Point

How can we proceed with discrepancy detection? The answer lies in leveraging what we already know about our data. This knowledge, or "data about data," is referred to as metadata.

Metadata provides the blueprint for what clean data should look like. By combining domain knowledge with the statistical insights from exploratory data analysis, analysts can systematically identify anomalies and inconsistencies.

Recall the techniques from Chapter 2: understanding data types, domains, acceptable value ranges, and statistical distributions gives us powerful tools for validation.

Key Questions

- What do I know about this data?
- What are the expected patterns?
- Where do values fall outside norms?

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Statistical Descriptions for Anomaly Detection

Basic statistical data descriptions help grasp data trends and identify anomalies that may indicate data quality issues. These foundational metrics provide objective criteria for flagging suspicious values.

01

Central Tendency

Calculate mean, median, and mode values to understand typical data points

02

Distribution Shape

Determine if data is symmetric or skewed, revealing potential sampling issues

03

Range Analysis

Identify minimum and maximum values, checking if all values fall within expected ranges

04

Dispersion Metrics

Compute standard deviation to measure spread and identify outliers beyond ±2σ from mean

05

Dependency Checks

Examine known relationships between attributes for violations of expected patterns

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Flagging Potential Outliers

Statistical Rule

Values that are more than two standard deviations away from the mean for a given attribute may be flagged as potential outliers requiring investigation.

Investigation Process

From statistical analysis, you may find noise, outliers, and unusual values that need investigation. Write custom scripts and use specialized tools to automate detection.

Analyst's Note: Be vigilant for inconsistent use of codes and any inconsistent data representations throughout the dataset. These subtle issues often escape automated detection.

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Special Error Patterns to Detect



Inconsistent Representations

Watch for mixed date formats like "2010/12/25" and "25/12/2010" in the same dataset



Field Overloading

Developers may squeeze new attribute definitions into unused portions of existing attributes, creating hidden data quality issues



Unique Rules

Each value of a given attribute must be different from all other values for that attribute



Consecutive Rules

No missing values allowed between lowest and highest values; all values must be unique (e.g., check numbers)

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Understanding Missing Values

As discussed in Section 3.2.1, understanding why values are missing helps determine appropriate handling strategies. Different causes require different approaches to imputation or exclusion.



Refusal or Non-Applicability

The person refuses to provide information or the attribute doesn't apply to them (e.g., license number left blank by nondrivers)



Unknown Correct Value

The data entry person lacks knowledge of the correct value and leaves the field blank



Deferred Value

The value is to be provided by a later step in the process and hasn't been populated yet

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Null Rules and Data Entry Conventions

Defining Null Conditions

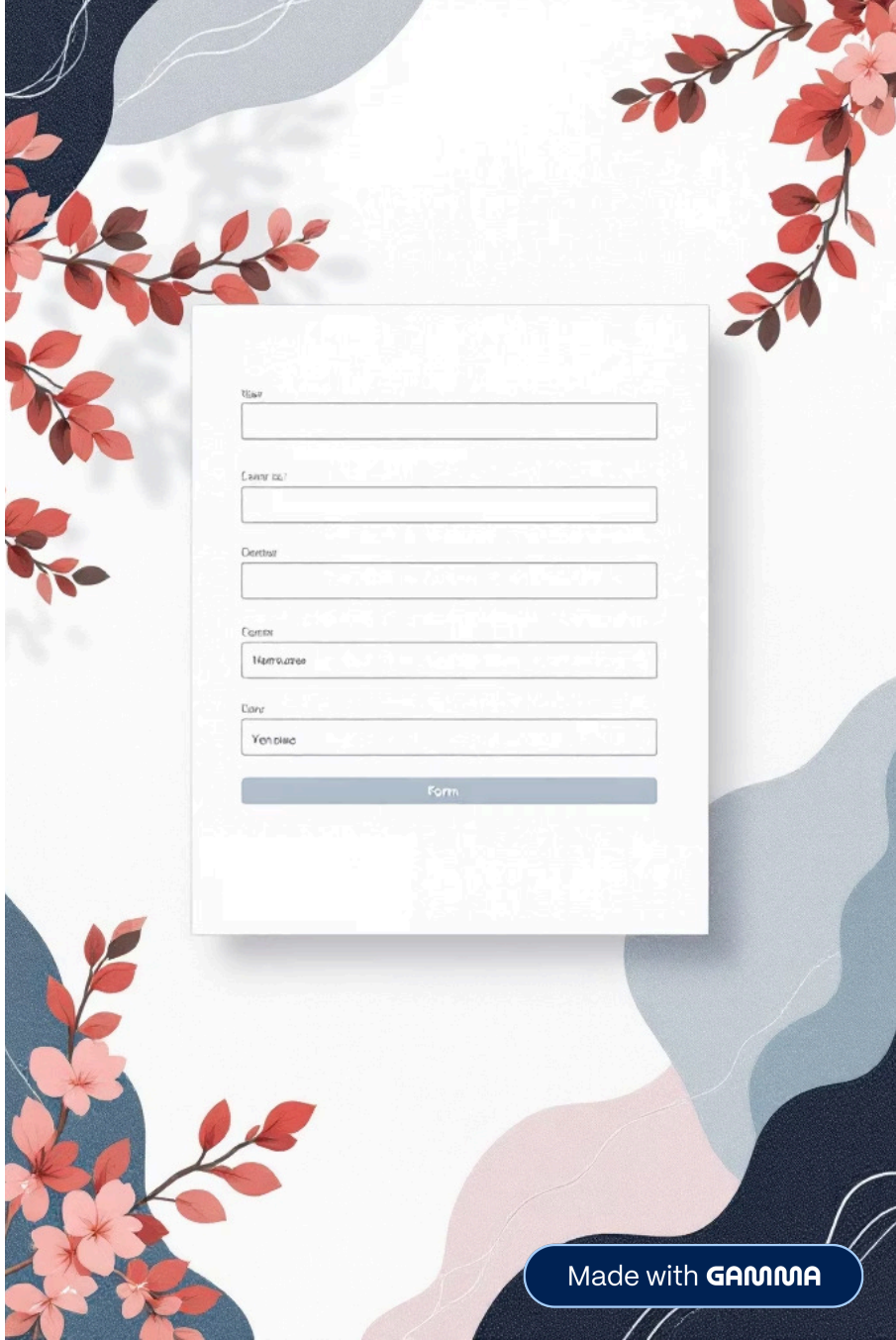
A null rule specifies the use of blanks, question marks, special characters, or other strings that may indicate the null condition (where a value for a given attribute is not available), and how such values should be handled.

Recording Conventions

The null rule should specify how to record the null condition:

- Store zero for numeric attributes
- Store blank for character attributes
- Transform entries like "don't know" or "?" to standardized blanks

Best Practice: Document your null handling conventions clearly so all team members process missing data consistently across the organization.



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